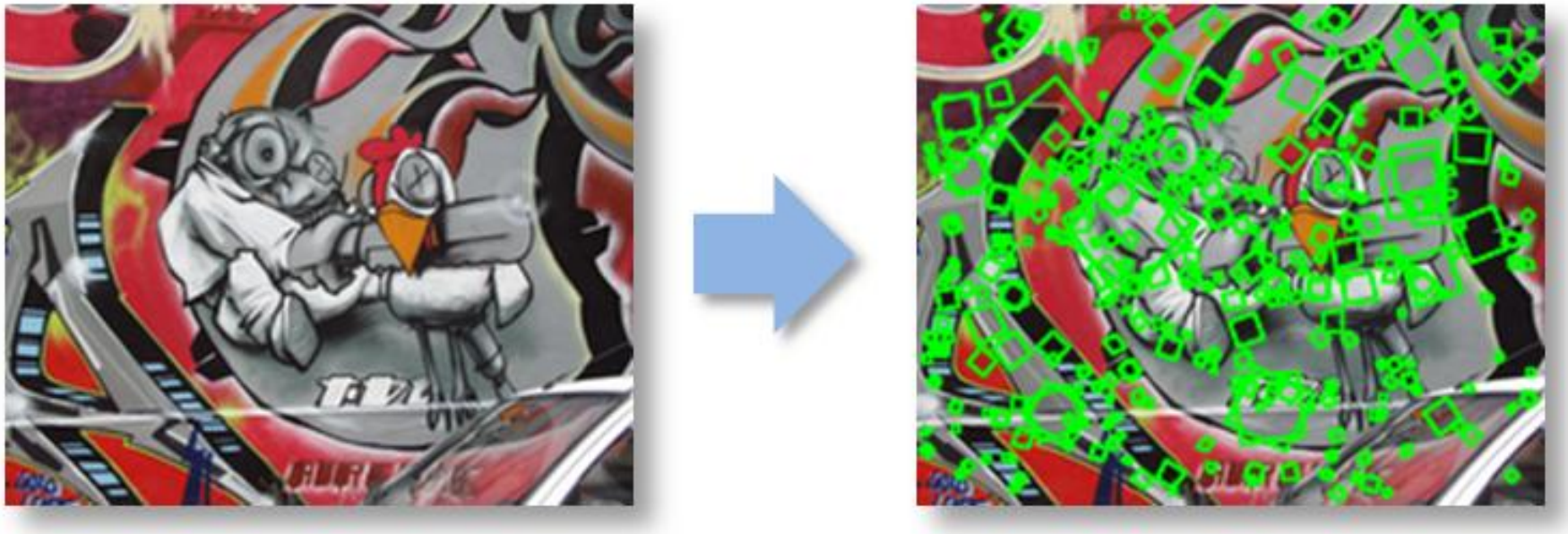
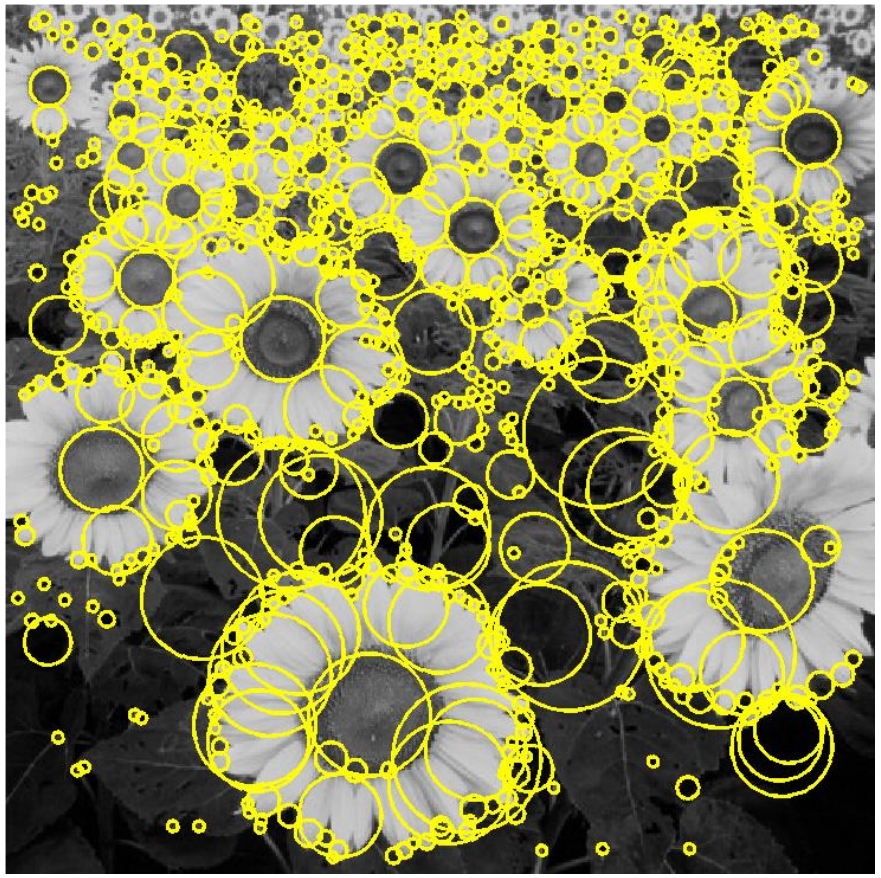
CS5187: Vision & Image

Local features & Harris corner detection

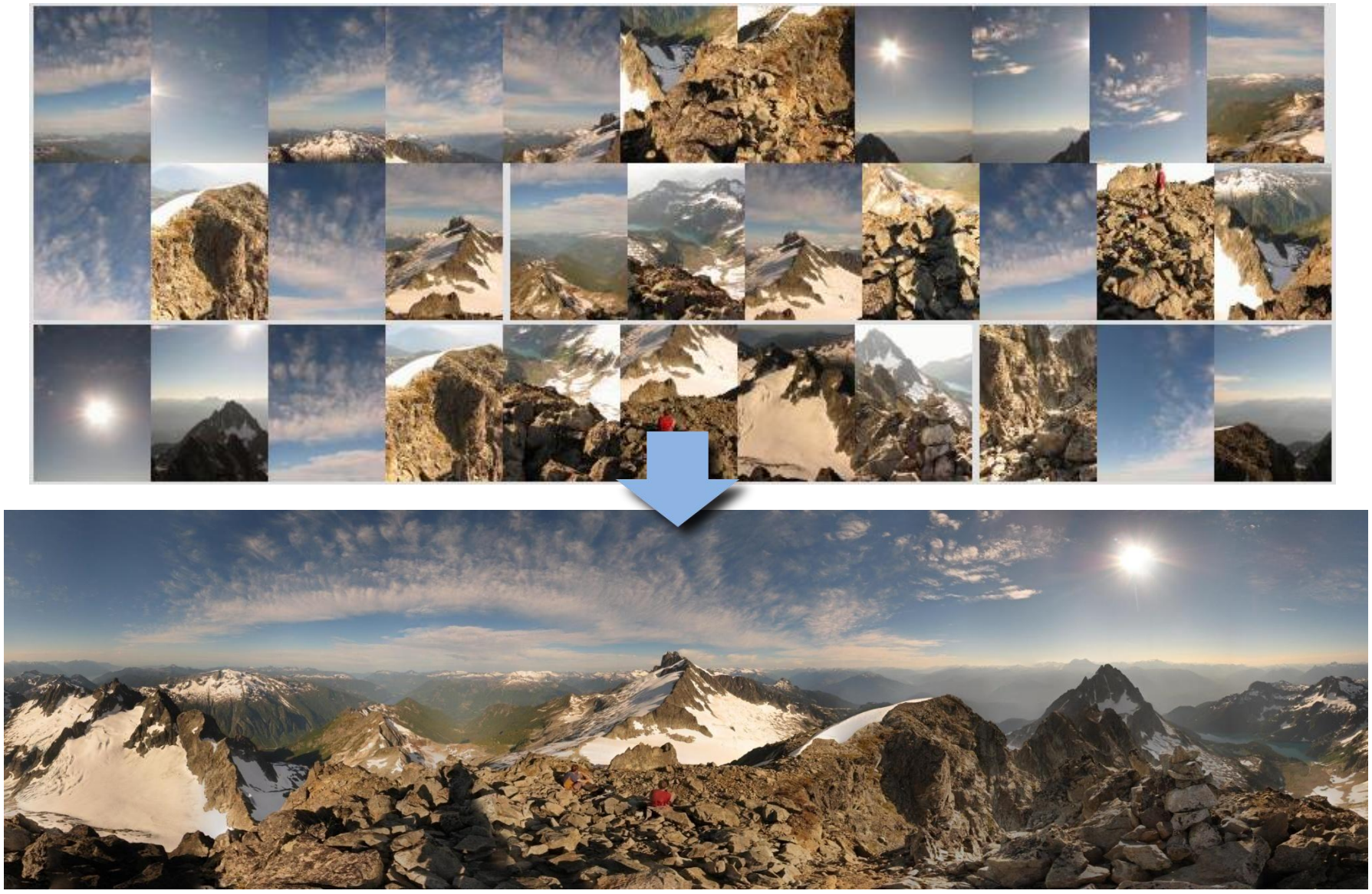


Today: Feature extraction—Corners and blobs





Motivation: Automatic panoramas



Credit: Matt Brown

Panorama stitching



Panorama captured by Perseverence Rover, Feb. 20, 2021

https://www.space.com/nasa-perseverance-rover-first-panorama-mars

Motivation: Automatic panoramas



GigaPan:
http://gigapan.com/

Also see Google Zoom Views:
https://www.google.com/culturalinstitute/beta/project/gigapixels

Why extract features?

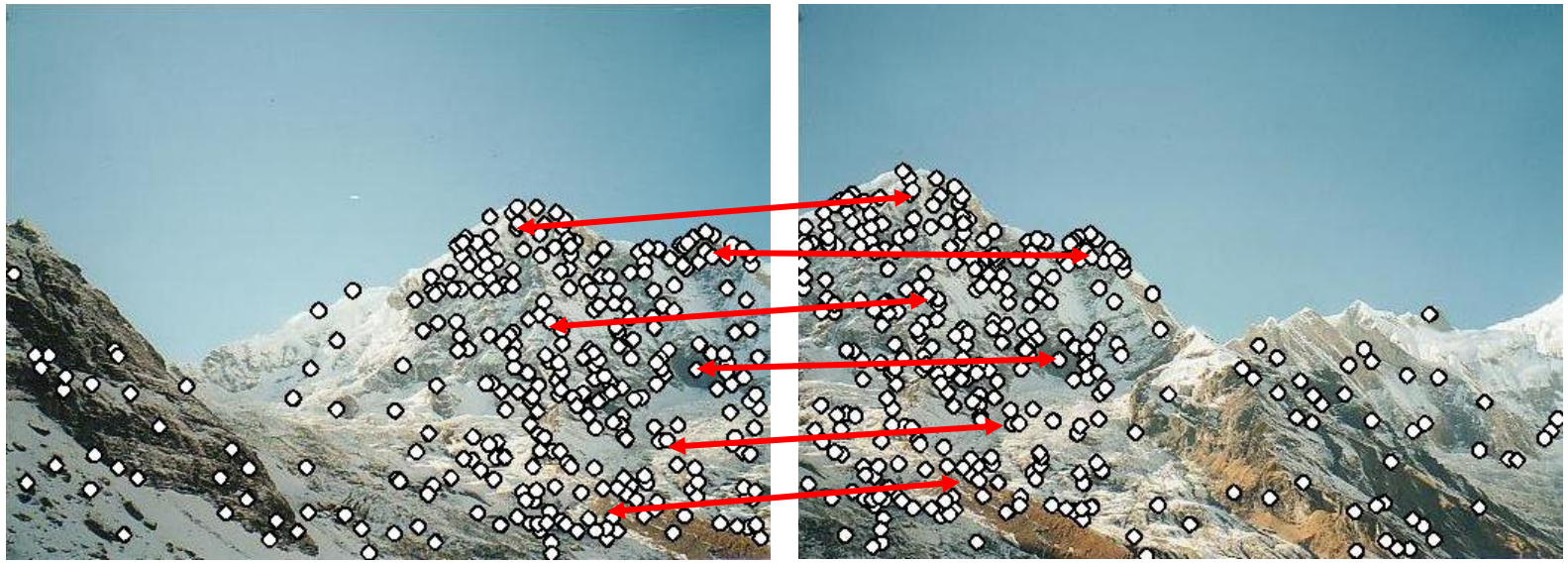
- Motivation: panorama stitching
  - We have two images – how do we combine them?





Why extract features?

- Motivation: panorama stitching
  - We have two images – how do we combine them?



Step 1: extract features
Step 2: match features

Why extract features?

- Motivation: panorama stitching
  - We have two images – how do we combine them?



Step 1: extract features
Step 2: match features
Step 3: align images